Question:
Carlos is planting a lemon tree. The tree will cost $90 to plant. Each year it will grow 7 lemons, which he can sell for $1.5 each. It costs $3 a year to water and feed the tree. How many years will it take before he starts earning money on the lemon tree?

Reasoning + Answer:

We are given:

- Cost to plant the tree: $90 (one-time cost)
- Each year, the tree produces 7 lemons
- Each lemon sells for $1.50
- Annual cost to water and feed the tree: $3

We need to find **how many years it will take before Carlos starts earning money** âĢĶ that is, when the **revenue from lemons exceeds the total costs (initial planting + annual expenses)**.

LetâĢĻs define:
- Let y be the number of years after planting.
- Revenue in year y: $7 \times 1.5 = 10.5$ dollars per year
- Annual cost: $3
- Net income per year after the initial cost: $10.5 - 3 = 7.5$ dollars per year (after accounting for water and feed)

But the **initial $90 cost** must be recovered before Carlos starts earning money.

So, we need to find the smallest integer y such that the **total net income** over y years is at least $90.

Total net income after y years:
$$
\text{Total income} = (10.5 - 3) \times y = 7.5y
$$

We want:
$$
7.5y \geq 90
$$

Solve for y:
$$
y \geq \frac{90}{7.5} = 12
$$

So, after **12 years**, Carlos will have earned back the initial cost.

Therefore, **after 12 years**, he starts earning money (since the cost is fully recovered, and from year 13 onward, he starts making profit).

âĿĵ But the question is: *How many years will it take before he starts earning money?*

That means: **when does the net profit become positive?**

Wait âĢĶ actually, "starts earning money" means when the **profit becomes positive**, i.e., when the **total income exceeds the total cost**.

So, we need to find the **first year** when the **net profit is positive**.

But the net profit is the cumulative profit after y years.

We want to find the smallest y such that:
$$
\text{Total revenue} - \text{Total cost} > 0
$$
$$
(10.5 \times y) - (90 + 3y) > 0
$$
$$
10.5y - 90 - 3y > 0
$$
$$
(10.5 - 3)y - 90 > 0
$$
$$
7.5y - 90 > 0
$$
$$
7.5y > 90
$$
$$
y > 12
$$

So, the total profit is positive **after 12 years âĢĶ meaning that **at the end of year 12**, the profit is exactly zero (break-even), and **after that**, in year 13, it becomes positive.

Therefore, Carlos **starts earning money** in year **13**.

But does "start earning money" mean when the net profit becomes positive?

Yes âĢĶ so after year 12, the profit is zero (break-even). In year 13, he earns more than he spends, so he starts earning money.

âľħ Final answer: **13 years**

LetâĢĻs verify:

- After 12 years:
- Revenue: $12 \times 7 \times 1.5 = 12 \times 10.5 = 126$
- Costs: $90 + 12 \times 3 = 90 + 36 = 126$
âĿĮ Break-even

- After 13 years: